Auteur: Michael Livchits
Studierichting: Informatica
Oefeningenreeks 1C nr. 31.15

$$\left(\frac{-3-5i}{i+4}\right)^5$$

$\frac{-3-5i}{i+4}$ vereenvoudigen:

$$\frac{-3-5i}{i+4}\cdot\frac{i-4}{i-4}$$

$$=-1-i$$

$$\Rightarrow(-1-i)^5$$

$$r=\sqrt{(-1)^2+(-1)^2}=\sqrt{2}$$

$$\sin\theta=\frac{-1}{\sqrt{2}}$$

$$\cos\theta=\frac{-1}{\sqrt{2}}$$

Derde kwadrant:

$$\theta=\frac{5\pi}{4}$$

$$\Rightarrow z=\left(\sqrt{2}\right)^5\cdot\left(\cos\left(\frac{25\pi}{4}\right)\sin\left(\frac{25\pi}{4}\right)\right)$$

$$=\left(\sqrt{2}\right)^5\cdot\left(\cos\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{4}\right)\right)$$

$$=4\sqrt{2}\cdot\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}i}{2}\right)$$

$$=4+4i$$

References